

Path Analysis

Steps of SEM

- 1. Model specification**

Translate your hypothesis into a model

- 2. Identification**

Make sure your model is mathematically estimable

- 3. Parameter Estimation**

Fit your model with software

- 4. Model Evaluation**

Assess how well your model fits the observed sample data/covariance

- 5. Model Modification**

Evaluate potential modifications to the model

Specification

The process of translating theory into equations

- Substantive theory (from your research field)
- Measurement theory (our general understanding of how measurement works)
- Experimental design (implied information from design)
 - Randomized variables are independent of other exogenous variables
 - Longitudinal data has certain implied structures

Specification Error: When an assumption of the model is incorrect (AKA misspecification)

e.g., Assume two variables are independent when they're not

e.g., Assume X predicts Y , when the reverse is true

These are unavoidable, focus on impact on the model

Path Diagrams

Path diagrams are a pictorial representation of a system of simultaneous equations

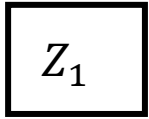
There should be a **one-to-one** correspondence between path diagram and equation

Often path diagrams are more intuitive than equations

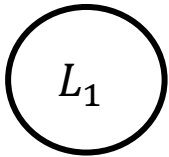
Process:

1. Draw an arrow from each predictor to its outcome
2. Two headed arrows among all exogenous variable
 - a. This is the typical assumption in path analysis but not SEM
3. Add disturbance/error to all outcomes (i.e., endogenous variables)
 - a. In path analysis, disturbance has it's own path coefficient. In SEM typically fixed to 1.
 - b. Disturbances are typically uncorrelated with each other and exogenous variables

Elements of a Path Diagram



Rectangle/Square: Observed variable



Circle/Ellipse: latent variable

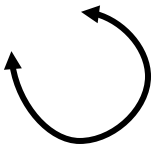
ϵ_1

Unenclosed variable: error/deviance

Note: sometimes in circle/ellipse

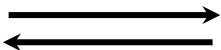


Single headed arrow: Variable at base predicts variable at head



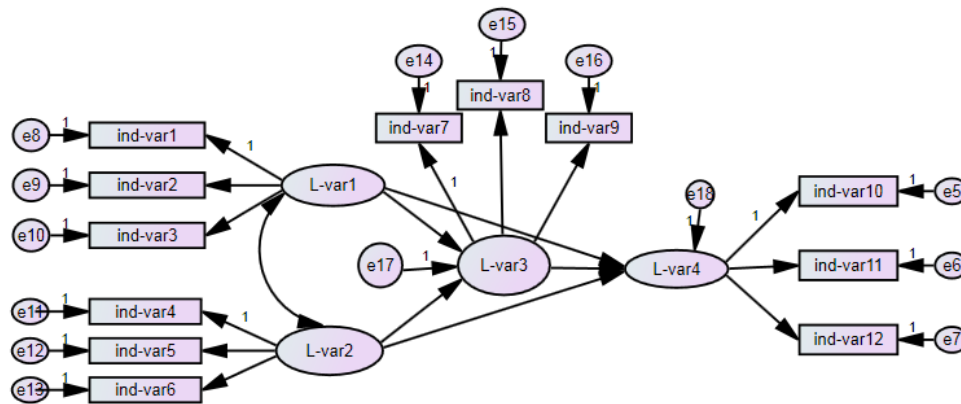
Two headed arrow: Variables at the ends are associated/correlated/allowed to covary

Two headed arrow at same variable: variance



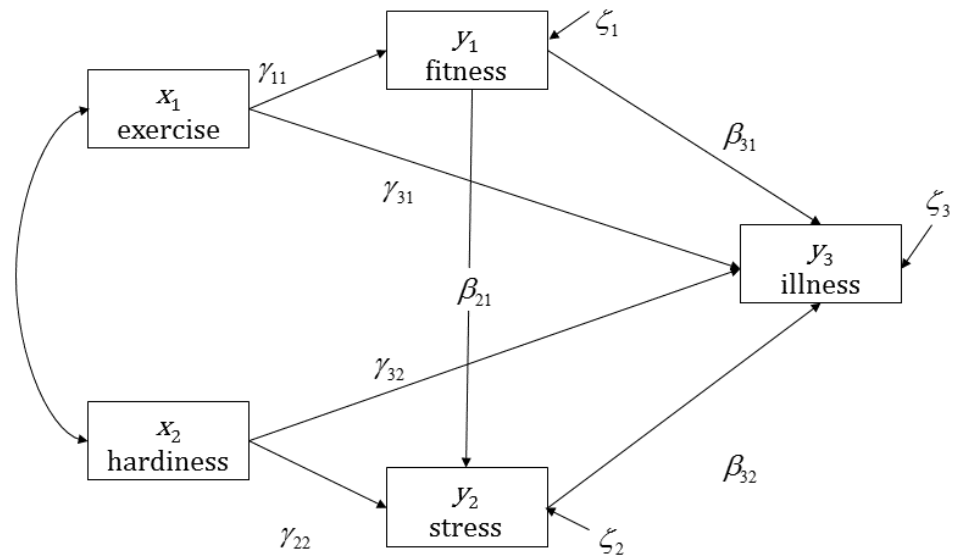
Bi-directional arrows: Feedback relationship

Examples of Path Diagrams in the Wild



A lack of arrows is informative!
No direct covariance

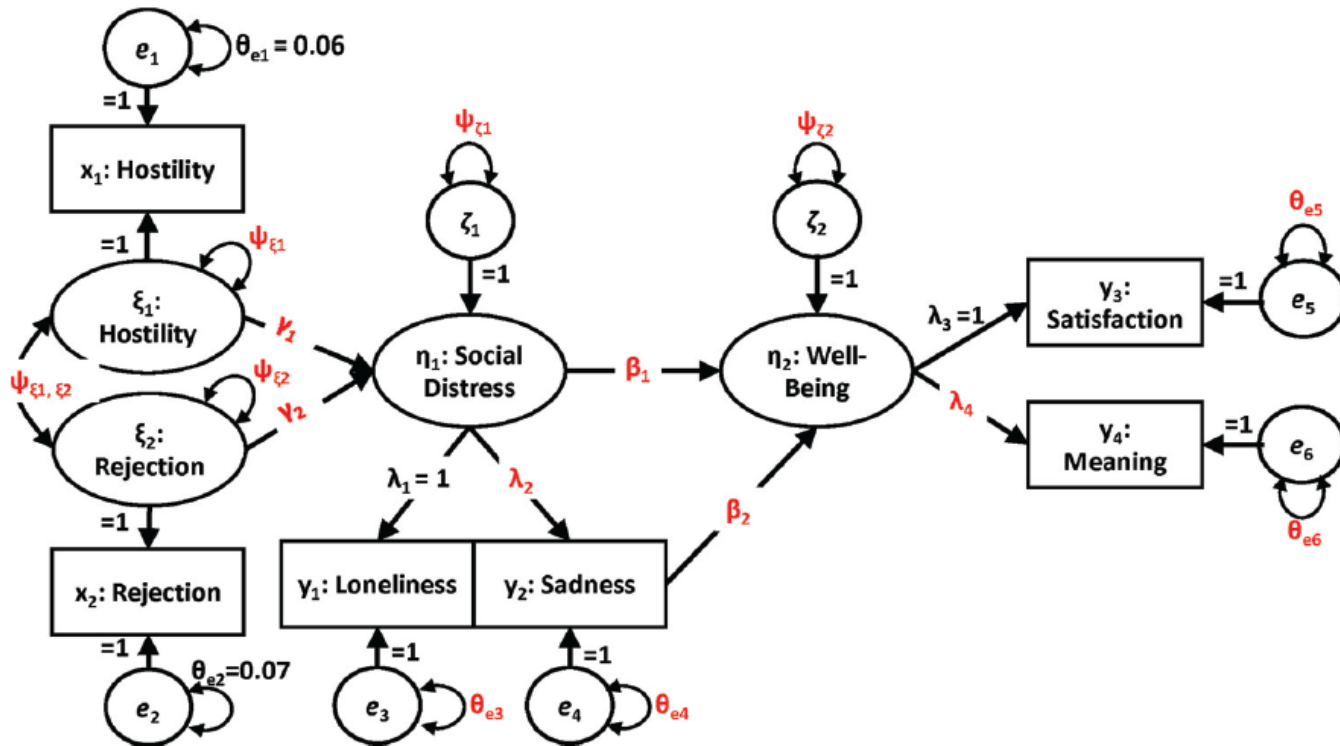
No variances typically means variables are standardized (i.e., exogenous variables have variance of 1)



A GREAT example!

Figure 18

Hypothesized Model of the Effect of Social Distress on Emotional Well-Being With the NIH Toolbox Norming Data Set



Note. In this model, perceived rejection and perceived hostility are treated as causal indicators of the social distress latent factor. Estimated parameters have red labels. See the online article for the color version of this figure.

Fife, Brunwasser, Merkle, 2022 *Psychological Methods*

Activity

Problem 1. Consider a research study where we are measuring peer support and anxiety over two time points. Peer support and anxiety are both latent variables each with three observed indicators at each time point.

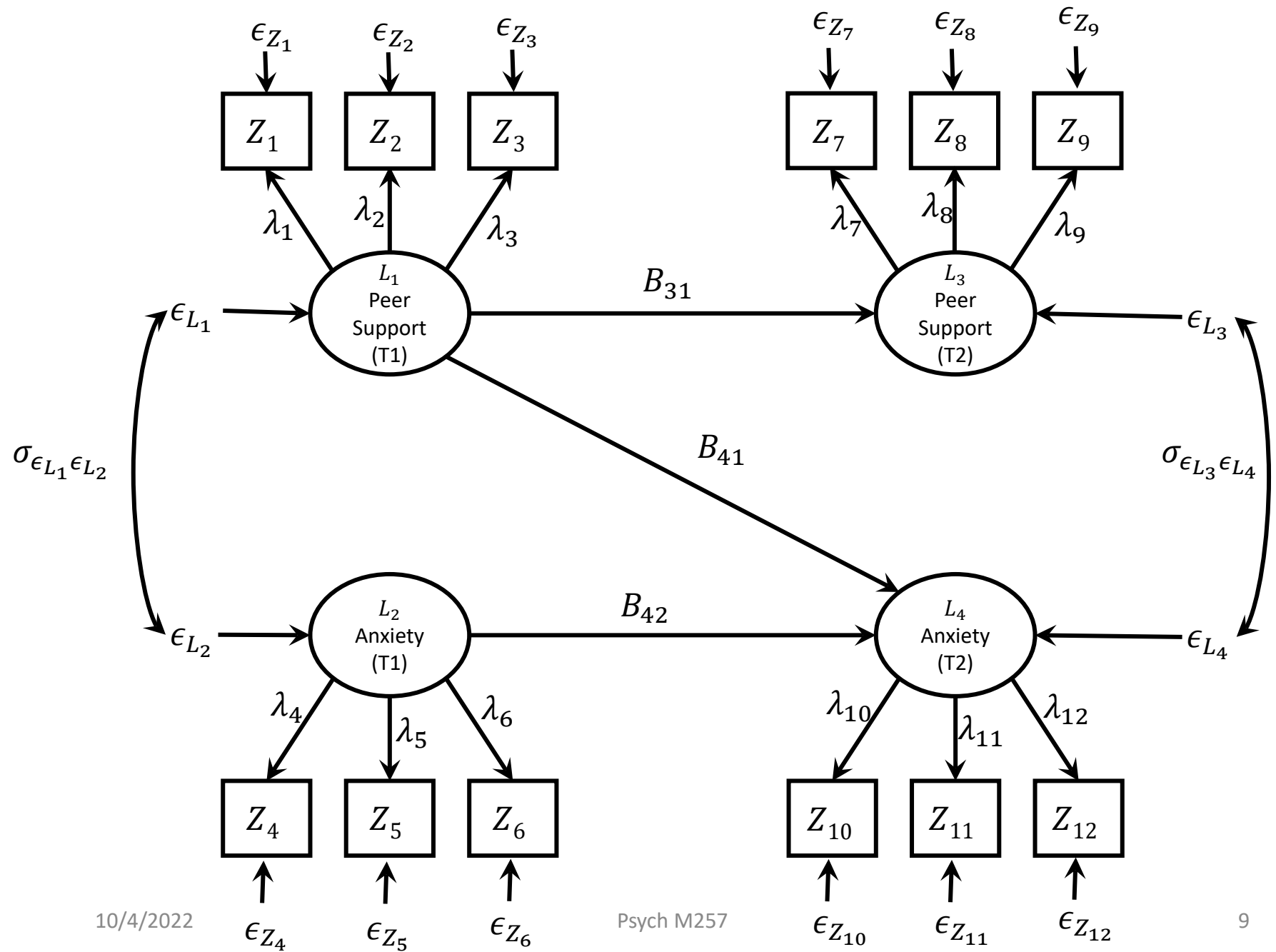
- a) Draw a path diagram which reflects the following hypothesis: Peer support and anxiety at time 1 predict anxiety at time 2. Peer support at time 1 also predicts peer support at time 2. Peer support and anxiety at time 2 do not have an assumed causal relationship, however their errors should be allowed to be correlated. Peer support and anxiety at time should be allowed to be correlated.

Make sure to include all your errors

You do not need to include means (intercepts) or variances

- b) Add notation to your path diagram

End Part a)



Activity

c) Based on your path diagram write out the individual equations for your latent variables. Write out the equation for one observed variable per latent variable.

$$\begin{aligned}L_{1i} &= \alpha_{L_1} + \epsilon_{L_{1i}} \\L_{2i} &= \alpha_{L_2} + \epsilon_{L_{2i}} \\L_{3i} &= \alpha_{L_3} + B_{31}L_{1i} + \epsilon_{L_{3i}} \\L_{4i} &= \alpha_{L_4} + B_{41}L_{1i} + B_{42}L_{2i} + \epsilon_{L_{4i}}\end{aligned}$$

$$Z_{1i} = \lambda_{1,1}L_{1i} + \epsilon_{Z_{1i}}$$

$$\dots \\Z_{4i} = \lambda_{4,2}L_{2i} + \epsilon_{Z_{4i}}$$

$$\dots \\Z_{7i} = \lambda_{7,3}L_{3i} + \epsilon_{Z_{7i}}$$

$$\dots \\Z_{10i} = \lambda_{10,4}L_{4i} + \epsilon_{Z_{10i}}$$

...

Matrices

d) Based on your path diagram write out the following matrices. Fill in their dimensions.

$$\mathbf{B} =$$

$$\Sigma_{\epsilon_Z} =$$

$$\Sigma_{\epsilon_L} =$$

$$\mathbf{\Lambda} =$$

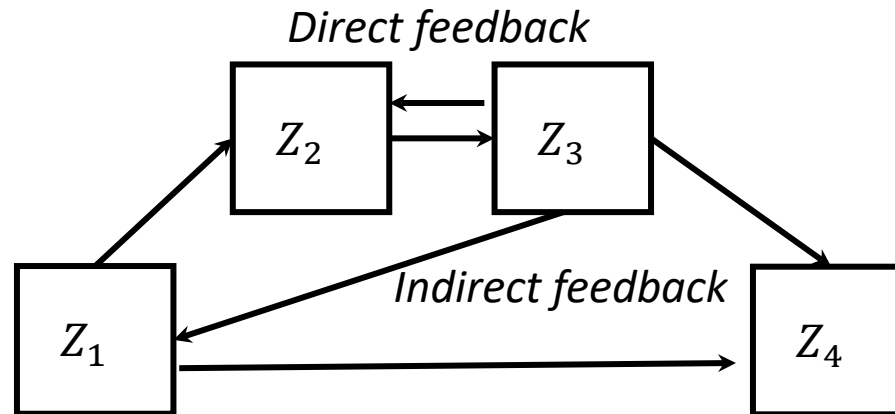
Implied Covariance

- Ways to find it:
 - Covariance Algebra
 - First law
 - Path tracing
 - Matrix Algebra
- Why do we care?
 - Remember in the notation lecture there was no covariance matrix for L
 - We use what we know about covariance to calculate an implied covariance matrix
 - Implied covariances are a function of model parameters
 - We use sample estimates of covariance to estimate model parameters

Recursive/Non-Recursive Models

Non-recursive Model: Involves a feedback loop, direct or indirect
i.e., can trace from a variable back to self

Recursive Model: No feedback loop



We will only deal with recursive models in this class

Is my model recursive?

If we write out equations for all endogenous variables, can you stack them such that endogenous variable in each equation is not a cause in a previous equation.

Recursive

$$L_4 = \alpha_{L_1} + B_{43}L_3$$

$$L_5 = \alpha_{L_4} + B_{54}L_4 + B_{52}L_2$$

$$L_6 = \alpha_{L_6} + B_{64}L_4 + B_{65}L_5 + B_{61}L_1$$

Non-recursive

$$L_4 = \alpha_{L_1} + B_{45}L_5$$

$$L_5 = \alpha_{L_4} + B_{56}L_6 + B_{52}L_2$$

$$L_6 = \alpha_{L_6} + B_{64}L_4 + B_{61}L_1$$

B_{YY} matrix is an upper or lower triangular matrix

$$B_{YY} = \begin{bmatrix} 0 & 0 & 0 \\ B_{54} & 0 & 0 \\ B_{64} & B_{65} & 0 \end{bmatrix}$$

B_{YY} matrix is an **not** upper or lower triangular matrix

$$B = \begin{bmatrix} 0 & B_{45} & 0 \\ 0 & 0 & B_{56} \\ B_{64} & 0 & 0 \end{bmatrix}$$

Path Tracing Rules

The purpose of path tracing rules is to determine what the model implied covariance matrix is ($\Sigma(\theta)$)

Assumes all variables are standardized (can back calculate from there)

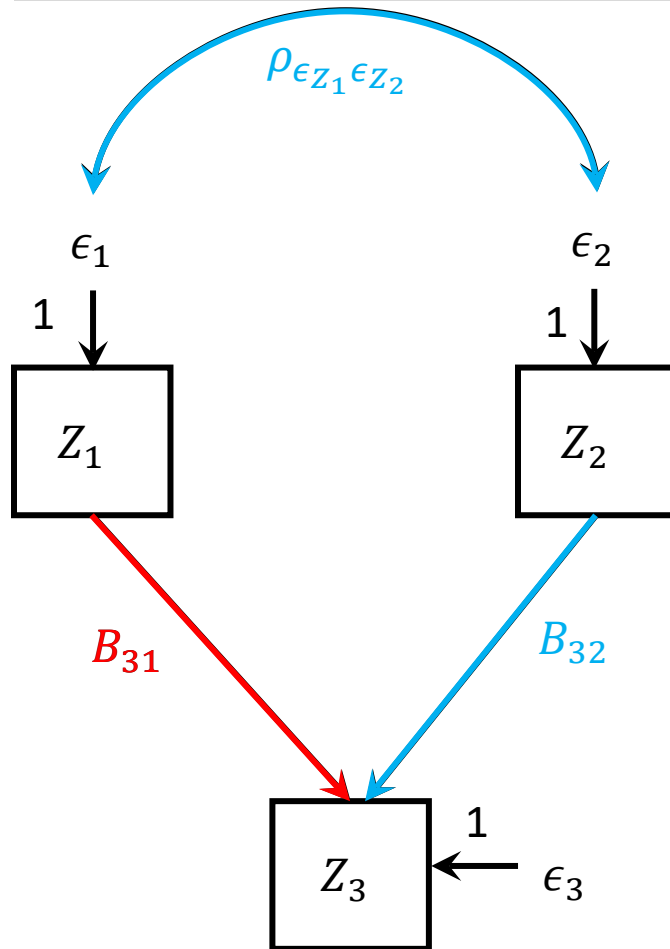
With variances added to path diagram rules generalize

Only works for recursive models (i.e., hierarchical models)

Correlation between Z_i and Z_j equals the sum of the product of all paths obtained from each possible tracing between Z_i and Z_j

- Do not enter the same variable twice
- Variables cannot be entered through an arrow head and left through an arrow head

Path Tracing Example



- Do not enter the same variable twice
- Variable cannot be entered through an arrow head and left through an arrow head

$$\rho_{Z_3Z_1} = B_{31} + \rho_{Z_1Z_2}B_{32}$$

$$\rho_{Z_3Z_2} = B_{32} + \rho_{Z_1Z_2}B_{31}$$

$$\rho_{Z_1Z_2} = \rho_{\epsilon_{Z_1}\epsilon_{Z_2}}$$

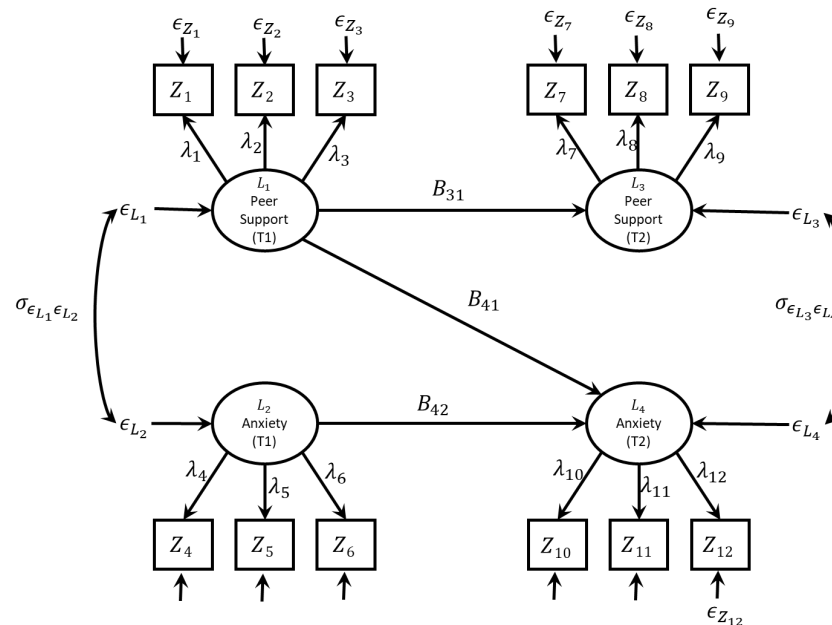
$\rho_{\epsilon_3Z_1} = 0$ The only path out of ϵ_3 is an arrow into Z_3 and the only path out of Z_3 is an arrow head

$\rho_{\epsilon_3\epsilon_2} = 0$

Path Tracing Activity

Use the path tracing rules to find the model implies correlation between L_3 and L_4

Can try to trace from L_3 to L_4 or L_4 to L_3 solutions are the same

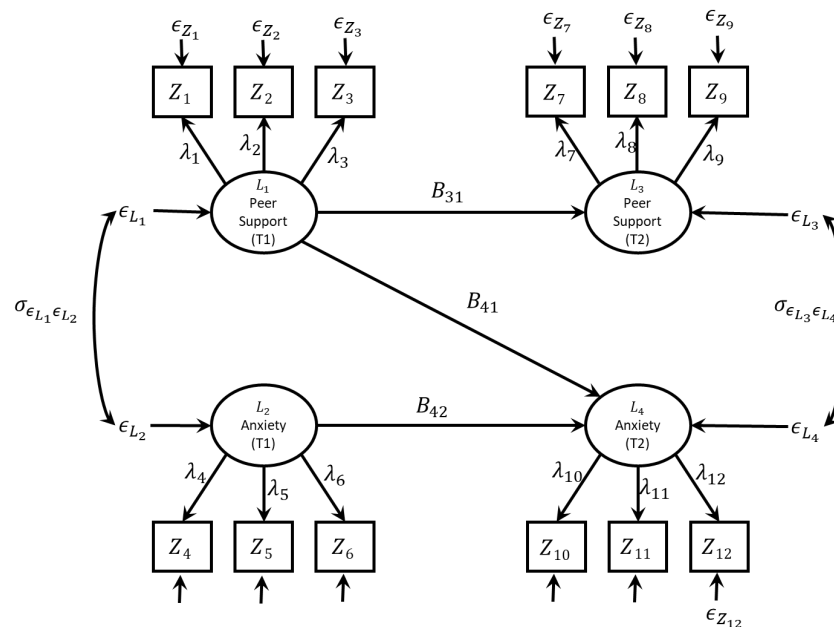


$$\rho_{L_3 L_4} = 1\sigma_{\epsilon_{L_3}\epsilon_{L_4}} + B_{31}B_{41} + B_{31}1\sigma_{\epsilon_{L_1}\epsilon_{L_2}}1B_{42}$$

Path Tracing Activity

Use the path tracing rules to find the correlation between Z_1 and Z_7

Try to trace from Z_1 to Z_7 or Z_7 to Z_1 solutions are the same



$$r_{Z_1 Z_7} = \lambda_1 B_{11} \lambda_7$$

All other paths go through L_4 which can't be left through an arrow head